The Impact of the COVID-19 Pandemic on the Chinese Economy: With a Look at the Pharmaceutical, Leisure, and Retail industries

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Introduction

The 2020 outbreak of the COVID-19 virus has played a significant worldwide impact both in regard to economy and other aspects. There is still argument about when and where the virus originated, but the first recognized pandemic outbreak happened in Wuhan, China starting in January 2020. The city experienced a lock down from January 23, 2020 to April 8, 2020. The virus then began to spread quickly in multiple countries, including the United States, Japan, France, and Italy, in March 2020.

Under this severe situation, China quickly responded to contain this pandemic nationwide by strict mobility policies, including long time, large-scale regional lock downs, and government macro policies to limit the economic shock. The Chinese economy did initially suffer a severe contraction, but the government response led to a relatively rapid turnaround that is still in progress. This report examines the pharmaceutical, leisure, and retail industries in China, including a comparison with macro economy and how these three sectors reacted differently to the pandemic. The report mainly analyzes the trends of GDP and its components, including final consumption, gross capital formation, and net exports, per capita consumption, and the three industries' operating incomes and stock prices in China from 2019 Q4 to 2020 Q3. The interestingly low correlation between the pharmaceutical industry stock prices and its operating incomes as well as the leisure industry's strong consistency with its stock prices and per capita consumption are noted and examined.

Data Collection

I used data from the Chinese government website, Chinese National Bureau of Statistics, and Wind terminals to complete this study. The Chinese government website provided Chinese quarterly GDP and per capita consumption from 2019 Q4 to 2020 Q3. The Chinese National Bureau of Statistics provided quarterly final consumption, gross capital formation, and net exports as percentage components of GDP from 2019 Q4 to 2020 Q3. The Wind terminals provided quarterly operating incomes and daily stock prices from 2019 Q4 to 2020 Q3 for the Chinese pharmaceutical, leisure, and retail sectors. The stock prices were converted to a quarterly basis by taking the average of the closing prices for the quarter. I ran three groups of correlations: correlations between industry operating incomes and GDP, between operating incomes and per capita consumption, and between operating incomes and industry stock prices. All data were converted to quarter-on-quarter percentage changes when running correlations, as I would like to track the short-term fluctuations in the economy.

Results

A. Macro Economy

The quarterly percentage changes of GDP from 2019 Q4 to 2020 Q1, 2020 Q1 to 2020 Q2, and 2020 Q2 to 2020 Q3 were -25.72%, 21.12%, and 6.42%, respectively. The quarterly percentage changes of per capita consumption in the same period were -16.62%, -20.58%, and 28.96%. Clearly, the trend of per capita consumption was different from that of GDP, that it continued to experience a loss until 2020 Q3 and rebounded slowly from 2020 Q3 to 2020 Q4.

To examine the divergence between GDP and personal consumption, we can first look at this formula:

GDP = Consumption + Investment + Government Spending + Net Exports

In 2020 Q2, gross capital formation's contribution rate to GDP, which includes enterprise and government investment, was 156.70%, and net exports' contribution rate to GDP was 16.60%. Compared to that of final consumption, -73.3%, investment and net exports clearly drove up GDP and offset the influence of consumption. Specific government subsidy policies are included here. On March 31, 2020, "the government announced new monetary policies, including the prospect of lower deposit reserve requirements for small banks in the future and an additional $140 billion for small banks to spur lending to small businesses." On April 3, 2020, "PBOC announced reserve requirement ratio (RRR) cuts for rural banks and some city commercial banks totaling 100 basis points, which was expected to release around $57 billion funds into the banking system." On June 17, 2020, "China's State Council called on banks to sacrifice $212 billion in profits in 2020 to finance cheap loans to companies to offset the economic fallout from the pandemic" (Tang, 2020).

The GDP growth was also encouraged by export demands that "the world [was] grappled with severe coronavirus outbreaks, driving up demand for medical gear and lockdown goods made in China." "China's trade surplus surged to $75.42 billion in October, at the fastest pace in almost three years" (Bermingham, & Leng, 2020).

Additionally, the International Monetary Fund's report of the China government spending, which "is forecasted to be $5,6878 billion in Dec 2020," compared to "$5,172 billion in Dec 2019," is a strong evidence that government spending also positively contributed to GDP (International Monetary Fund, 2020).

We could clearly see that with increasing investment, government spending, and net export, which stimulated GDP from 2020 Q1 to Q2, the gap between GDP and consumption in the first half of year 2020 was explainable.

B. Responses of the three sectors

We can first look at the industry operating incomes.

From 2019 Q4 to 2020 Q1, all three industries experienced a decline revealed by the percentage changes of their operating incomes, which were -12.32%, -51.44%, -26.49%. The leisure and retail sectors experienced a larger loss compared to the pharmaceutical sector, as most of expenditures were by consumers and were discretionary, and mobility restriction policies caused people to postpone spending on "all sub-sectors of the leisure industry, including vacations, hotels, catering, and travels" (Liu, 2020). From 2020 Q1 to 2020 Q2, the three industries began to rebound, while the pharmaceutical industry again was the fastest to recover with a 21.04% percentage growth, being ahead of the other two industries. From 2020 Q2 to 2020 Q3, all three sectors were back to normal, especially the leisure sector that the quarterly percentage change of its operating income was 54.92%, indicating that once the economy rebounded, people had the ability and desire to pick up their spending.

Overall, the correlations between the pharmaceutical industry and GDP as well as that between the retail industry and GDP were as high as 0.98, showing that both sectors were consistent with GDP and experienced a similar fluctuation trend. The leisure industry had the biggest drop in 2020 Q1, a small rebound in Q2, and a large increase in Q3, similar to the trend of per capita consumption. We could firmly prove the tie between national consumption overall and leisure spending by the correlation between the leisure industry operating incomes and per capita consumption, which was 0.76, the highest among three correlation numbers between industry operating incomes and per capita consumption. The pharmaceutical industry acted differently from the leisure industry, as it only experienced a little loss and came back quickly, with reasons that will be further explained in the later paragraphs. The year-on-year percentage

changes of retail and leisure industries' operating incomes from 2019 Q3 to 2020 Q3 were -12.50% and -4.95%, which were both negative, indicating that the pandemic had a significant influence on these two industries and that they were not fully recovered yet. The correlations between three industries' operating incomes and GDP were all positive, indicating that all sectors were procyclical.

We can then look at the industry stock prices.

From 2019 Q4 to 2020 Q1, the decline of the stock market was not as much as that of the industry output at all. Only the leisure industry stock prices drooped a little by 4.89%. From 2020 Q1 to 2020 Q2, the pharmaceutical industry stock prices again rose more than those of the other two sectors by 12.01%. From 2020 Q2 to 2020 Q3, the rebound in the leisure industry stock prices was particularly strong with a 44.81% percentage growth, showing that once it became clear that the pandemic had been under control and would soon pass, investors would bid up the stock prices.

The stock prices did not show more of a decline as the operating incomes did, quite likely because China's central bank indirectly supported the stock market by keeping credit cheap and plentiful. The national government "injected $174 billion worth of liquidity into the market through reverse repo operations" and "adopted more flexible monetary policy measures to alleviate the impact of the epidemic" (Liu, & Yap, 2020).

We encountered another aspect where the pharmaceutical sector behaved differently from the other two sectors that the correlation between its industry operating incomes and stock prices was only 0.27, the lowest correlation number among three sectors. There were three possible reasons. First, investors might have been looking ahead in the future with confidence based on the pharmaceutical industry's great performance in 2019. Second, there are many factors that

determine the pharmaceutical industry's development. "The long-term trend of the pharmaceutical industry was not affected by the pandemic, and the pandemic brought more structural opportunities to related concept stocks, so the long-term value of the enterprise was not affected" (Wang, 2020). Third, people's short-term demands for medical treatments over the pandemic, including vaccines, blood products, CXO, bulk drug, and medical consumables, also boosted the industry stock prices. For instance, "Wuhan had been doing a large-scale purchase, including non-over-evaluated drugs and insulin preparations" in January because of the pandemic outbreak (Li, 2020). Overall, the investors were probably more optimistic with pharmaceutical, holding on and even purchasing the stocks, and thus did not let its stock prices suffered a decline.

On the contrary, the correlation between the leisure industry operating incomes and its stock prices was 0.94, the highest among three sectors, showing that the performance of the leisure industry stock prices was the most consistent with the industry operating income. The reason that the leisure industry was always consistent with the national economy might be that people tend to spend on leisure activities when the economy is doing well. The performance of the leisure industry more depends on the current situation, unlike the pharmaceutical industry, which also depends on a forecast of the future.

Conclusion

The components of GDP, including consumption, investment, and exports, as well as the pharmaceutical, leisure, and retail industries all had different responses to the COVID-19 pandemic in 2020. Macroeconomically, government investment and spending as well as net exports surged with implementations of national policies starting in late Q1 in 2020, which

boosted GDP and the stock market. Personal consumption diverged from this increasing trend in 2020 Q2 because of citizens' limited activities and spending. It then grew with other macroeconomic indicators in Q3. For the pharmaceutical, leisure, and retail sectors, the pharmaceutical industry had a better performance than the leisure and retail industries in 2020 Q1 and Q2, then all three sectors went back to normal in Q3, looking at their operating incomes and stock prices. The reason that the pharmaceutical industry acted differently from the other two sectors, and that the leisure industry is very consistent with national economy, might be that investors were more optimistic with the pharmaceutical industry, whose growth depend on many factors other than the the pandemic, and people's spending, which is usually reflected by the performances of the leisure industry, more depends on the performance of current economy rather than a future forecast.

Appendix

Table 1: Data about macro economy: GDP, GDP components, and per capita consumption

	year	2019	2020		
	quarter	4	1	2	3
GDP	billion yuan	27801.97	20650.43	25011.01	26617.20
	% increase (Quarter to Quarter)		-25.72	21.12	6.42
Per Capita Consumption	yuan	6095.00	5082.00	4036.00	5205.00
	% increase (Quarter to Quarter)		-16.62	-20.58	28.96
Final Consumption's Contribution Rate to GDP (Consumption + Government Spending)	%	57.90	64.10	-73.30	34.90
	% increase (Quarter to Quarter)		0.11	-2.14	-1.48
Gross Capital Formation's Contribution Rate to GDP (Enterprise + Government Investment)	%	44.30	21.50	156.70	52.00
	% increase (Quarter to Quarter)		-51.47	628.84	-66.82
Net Exports' Contribution Rate to GDP	%	-2.20	14.40	16.60	13.10
	% increase (Quarter to Quarter)		-754.55	15.28	-21.08

Table 2: Data about the three industries: operating incomes and stock prices

	year	2019	2020		
	quarter	4	1	2	3
operating income (billion yuan)	pharmaceutical industry	572.44	501.90	607.52	637.63
	% increase (Quarter to Quarter)		-12.32	21.04	4.96
	leisure industry	35.69	17.33	19.43	30.10
	% increase (Quarter to Quarter)		-51.44	12.12	54.92
	retail industry	342.37	251.69	276.21	291.88
	% increase (Quarter to Quarter)		-26.49	9.74	5.67
stock price (average of the closing price for the entire quarter) (yuan)	pharmaceutical industry	1989.45	2144.42	2401.89	2975.34
	% increase (Quarter to Quarter)		7.79	12.01	23.87
	leisure industry	1225.63	1165.64	1272.98	1843.44
	% increase (Quarter to Quarter)		-4.89	9.21	44.81
	retail industry	1316.81	1333.89	1402.36	1687.47
	% increase (Quarter to Quarter)		1.30	5.13	20.33

Table 3: Correlations

Correlations

	per capita consumption	GDP	stock price
pharmaceutical income	-0.05	0.98	0.27
leisure income	0.76	0.75	0.94
retail income	0.34	0.98	0.58

Data Resources

Chinese Government Website, http://www.gov.cn/.

Chinese National Bureau of Statistics, http://www.stats.gov.cn/.

Wind Terminals

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